

1 Model Used: YOLO-Based Object Detection System

1.1 Overview

The vision system is based on a lightweight YOLO-style object detection model exported in ONNX format for real-time edge inference. The model is optimized for embedded systems and low-power devices, enabling fast inference while maintaining acceptable detection accuracy.

The deployed model follows a single-stage object detection pipeline, where bounding boxes and class probabilities are predicted directly from the input image without requiring separate region proposal stages.

1.2 Model Architecture

The model is based on a YOLO (You Only Look Once) architecture, which performs object detection in a single forward pass of the neural network. This design significantly reduces computational cost compared to traditional two-stage detectors.

The network consists of:

- Convolutional backbone for feature extraction
- Feature pyramid-based neck for multi-scale feature fusion
- Detection head for bounding box, objectness, and class prediction

The model is exported to ONNX (Open Neural Network Exchange) format to enable cross-platform deployment and hardware acceleration.

1.3 Inference Pipeline

Each input frame is processed through the following steps:

- Frame capture from camera
- Resize to 640×640
- Pixel normalization to floating-point format
- Channel-wise scaling (mean and standard deviation)
- ONNX inference execution
- Decoding of output tensor into bounding boxes and class labels

1.4 Post-Processing

The model outputs bounding boxes and class predictions. These are filtered using confidence thresholds.

Traditional YOLO systems use Non-Maximum Suppression (NMS) to remove duplicate detections. In this implementation, post-processing is optimized for reduced CPU overhead to improve real-time performance on embedded systems.

1.5 Deployment Format

The model is deployed in ONNX format (`.onnx`), which provides:

- Hardware-independent execution
- Compatibility with embedded platforms such as Raspberry Pi and ARM devices
- Fast inference using ONNX Runtime

1.6 References

- YOLO Documentation: <https://docs.ultralytics.com/models/yolo26>
- ONNX Model Used: <https://huggingface.co/zwh20081/yolo26-onnx/resolve/main/yolo26n.onnx>

The implemented model is based on the reference below, which provides complete technical details and documentation of the object detection architecture.